

MOMENTUM HELICITY SEPARATES CASCADE OPERATORS FROM EULER–ARNOLD OPERATORS

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ABSTRACT. Paper III leaves open whether a local cascade operator in Tao’s sense can be an Euler–Arnold operator, and hands the question to real algebraic geometry. The obstruction there is structural: every argument of Paper I–Paper III is built on the energy identity, which is precisely the invariant Tao’s cancellation condition (4.3) is designed to preserve, so no linear functional can separate the two classes (Paper II, Corollary 5.4). Three-dimensional ideal flow carries a second structural invariant that nothing in (4.3) imposes: the helicity of the *momentum* $m = Au$, a Casimir of the Lie–Poisson structure and therefore conserved by every Euler–Arnold flow, for every inertia operator. We show that the resulting condition collapses to linear algebra. On $k^\perp$ the curl acts as $i|k|J$ with J a quarter turn, $A^{1/2}$ is a symmetric 2×2 matrix, and $SJS = \det(S)J$ for symmetric S ; consequently the helicity defect is *linear in the numbers $\sqrt{\det A_k}$ and sees the inertia operator through nothing else*. An operator is Euler–Arnold only if an explicit matrix $M(C)$ has an entrywise positive null vector. Euler–Arnold vertices give corank exactly one, with that null vector; cascade operators give full rank, for two, three and four species, on five mode sets, and for Tao’s own coefficients the null vector exists but fails positivity. The species count never enters the criterion, which is why it reaches the case Paper III could not. Finally, the criterion is unchanged by viscosity: the dissipative term is common to both classes and cancels from the comparison, so the exclusion holds for the viscous problem uniformly in ν , and a cascade operator is seen to produce helicity at a rate that survives the inviscid limit. The same reduction bounds the method. Applied one step earlier, before the time derivative, it gives the *value* of the helicity and not only its conservation as a linear functional of those same determinants; the entire Casimir content of an inertia operator is therefore that one vector, and the fibres of the map carry two free parameters per leg pair, the ratio of the eigenvalues and the orientation of the principal axes inside $k^\perp$. Along them the criterion, its null vector and the conserved helicity are constant to machine precision, while the vortex-stretching production is not.

1. THE QUESTION, AND WHY THE TRILOGY COULD NOT ANSWER IT

Paper III ends at a dimension count. On a web of triads it measures two parametrised images, that of $\{A_k\} \mapsto (T_A(\tau))_\tau$ and that of $(\alpha, \{G_k\}) \mapsto ((G \otimes G \otimes G)\alpha)_\tau$, finds a codimension growing quadratically in the number of legs, and stops, because two subvarieties whose dimensions sum to less than the ambient may still meet in special position.

The obstruction to going further is not the dimension count. It is that Paper I–Paper III are built on one invariant, and it is the wrong one. The Euler–Arnold class conserves energy; Tao’s condition (4.3) is imposed *precisely so that his averaged operator does too* [10, §4]; and Paper II, Theorem 5.1 shows the energy relations exhaust the linear information available on a triad. Corollary 5.4 then records the consequence: no linear functional vanishing on the Euler–Arnold class can exclude any cascade operator. Three papers were spent searching for a separator among functionals that every cascade satisfies by construction.

Three-dimensional ideal flow has a second structural invariant. For a divergence-free momentum one-form m on $\mathbb{T}^3$ the helicity

$$H(m) = \int_{\mathbb{T}^3} m \wedge dm = \int_{\mathbb{T}^3} m \cdot \operatorname{curl} m$$

is a Casimir of the Lie–Poisson structure on $\mathfrak{sdiff}(\mathbb{T}^3)^*$; see [5, §I.9], though Lemma 3.1 below proves what we need directly and in three lines. Casimirs do not depend on the Hamiltonian, so $H(m)$ with $m = Au$ is conserved by *every* Euler–Arnold flow, whatever the inertia operator; and nothing in (4.3) imposes it on a cascade. That asymmetry is the content of this paper.

The condition it yields is not a linear functional on the vertex — it involves A — so Corollary 5.4 does not reach it. It does not use invertibility of the frames, and the species count does not appear in it, so the $m \geq 3$ wall of Paper III does not reach it either.

Remark 1.1 (relation to the literature). Tao states the deficiency himself, and locates it in the vorticity formulation: the true nonlinearity admits one involving only differential operators, whereas a general averaged operator does not, so that any proof of global regularity must use structure of the nonlinearity that the averaged version does not share [10]. What is left implicit is *which* structure. This paper names one, and the naming is what makes it testable — the vorticity formulation is what carries the Casimirs of the Lie–Poisson structure, and the momentum helicity is the scalar that survives of them.

The complementary programme is well developed, and it is worth saying where this sits relative to it. The α -models are designed to preserve exactly what Tao’s averaging discards: the Navier–Stokes- α model is obtained by filtering the velocity of the fluid loop in Kelvin’s circulation theorem [8], and the Lagrangian-averaged Euler equations are Euler–Poincaré equations on the same Lie algebra with a modified inertia operator [9] — that is, they are members of the class $\{B_A\}$ studied here, with $A = I - \alpha^2 \Delta$ in place of $A = I$. The two constructions bracket the question. Averaging that respects the circulation theorem stays inside the Euler–Arnold class by design; averaging of Tao’s local cascade type, as shown below, cannot enter it for any inertia operator whatsoever.

A third line attacks the same construction from the side of estimates rather than of invariants. There the relevant structure is that the strain is a Calderón–Zygmund singular integral of the vorticity, so that the two are not independent; Constantin and Fefferman [6] showed that coherence of the vorticity direction depletes the stretching term, and the geometric criteria descending from that work all exploit the same link. An averaged operator severs it, since its “vorticity” is no longer the curl of the velocity that stretches it. Kiriukhin [7] reports, in a Fourier–Galerkin truncation reduced by the octahedral group, an orbit-level subcriticality of the stretching which the true nonlinearity satisfies and which Tao’s averaged construction violates. We have not verified that argument, and record it only to observe that its conclusion agrees with ours while its route does not: two structural properties of unrelated type — one an exact invariant of the Lie–Poisson geometry, one an estimate on nonlinear transfer — both failing for the same construction. The criterion developed below is of the first kind, and says nothing about the second.

2. PRELIMINARIES

We keep the notation of Paper I. Let $\mathfrak{g}$ be a Lie algebra with an inner product $\langle \cdot, \cdot \rangle$ and $A: \mathfrak{g} \rightarrow \mathfrak{g}$ symmetric and positive definite, with ad^* defined by $\langle \operatorname{ad}_x^* y, z \rangle = \langle y, [x, z] \rangle$. In the energy-normalised variable $X = A^{-1/2} m$, the Euler–Arnold equation reads $\dot{X} = B_A(X, X)$ with

$$B_A(X, Y) = \frac{1}{2} \left(A^{-1/2} \operatorname{ad}_{A^{-1/2} X}^* (A^{1/2} Y) + A^{-1/2} \operatorname{ad}_{A^{-1/2} Y}^* (A^{1/2} X) \right), \quad (1)$$

and $S(x, y, z) := \langle y, [x, z] \rangle$. Taking $Y = X$ in (1) and writing

$$u := A^{-1/2}X, \quad m := A^{1/2}X, \quad \text{so that } m = Au, \quad (2)$$

gives $\langle B_A(X, X), Z \rangle = S(u, m, A^{-1/2}Z)$, that is

$$A^{1/2}B_A(X, X) = \text{ad}_u^* m = \dot{m}. \quad (3)$$

Throughout, $\mathfrak{g}$ is the algebra of divergence-free, mean-zero vector fields on $\mathbb{T}^3$, and we work on a finite set $\mathcal{S}$ of wavevectors closed under $k \mapsto -k$. Since the fields are real-valued, their Fourier coefficients satisfy $\hat{v}(-k) = \overline{\hat{v}(k)}$, and for A to preserve that we need $A_{-k} = A_k$ as operators on $k^\perp = (-k)^\perp$, so A contributes one symmetric 2×2 matrix per pair $\{k, -k\}$; we write $\mathcal{P}$ for the set of pairs.

Definition 2.1. A local cascade operator with m species is determined by constants $\alpha \in \mathbb{R}^{m^3}$ and divergence-free profiles $\psi_i(k) \in k^\perp$ through

$$\langle C(X, X), Z \rangle = \sum_{k+l+p=0} \sum_{i,j,q} \alpha_{ijq} \langle \psi_i(k), \hat{X}(k) \rangle \langle \psi_j(l), \hat{X}(l) \rangle \langle \psi_q(p), \hat{Z}(p) \rangle,$$

the sum over ordered triples from $\mathcal{S}$.

Lemma 2.2. Under Tao's symmetry (4.2), his cancellation condition (4.3) is exactly the vanishing of the cyclic symmetrisation of α ,

$$\alpha_{ijq} + \alpha_{jq i} + \alpha_{qi j} = 0 \quad \text{for all } i, j, q,$$

and this holds if and only if $\langle C(X, X), X \rangle \equiv 0$. Together with (4.2) itself the admissible α therefore form a linear space of dimension $\frac{1}{3}(m^3 - m)$, namely 2, 8, 20 for $m = 2, 3, 4$. Dropping (4.2) and imposing only the cyclic condition doubles this to $\frac{2}{3}(m^3 - m)$; the extra directions are those antisymmetric in the first two indices, and they contribute nothing, since the triad sum runs over ordered pairs and is therefore symmetric in those indices. They give the zero operator.

Proof. By (4.2), $\alpha_{i_1 i_2 i_3} = \alpha_{i_2 i_1 i_3}$, so the six permutations occurring in (4.3) coincide in pairs and their sum is twice the cyclic sum. Setting $Z = X$ in Definition 2.1 and grouping the ordered triples (k, l, p) into cyclic classes, the coefficient of $\langle \psi_i(k), \hat{X}(k) \rangle \langle \psi_j(l), \hat{X}(l) \rangle \langle \psi_q(p), \hat{X}(p) \rangle$ is $\alpha_{ijq} + \alpha_{jq i} + \alpha_{qi j}$; for this to vanish for all profiles and all X is the stated condition. The dimension count is m^3 less the number of cyclic orbits in $\{1, \dots, m\}^3$, which is $m + (m^3 - m)/3$. $\square$

Remark 2.3. Lemma 2.2 is Paper II, Theorem 5.1 read on the coefficients rather than on the blocks, and it is the constraint that was omitted from the measurement in Paper III, Proposition 8.1; the corrected table there now uses it.

3. THE CASIMIR

Lemma 3.1. For every A and every solution of the Euler–Arnold equation, $\frac{d}{dt}H(m) = 0$, where $m = A^{1/2}X$ as in (2).

Proof. It suffices that $\langle m, [\text{curl } m, w] \rangle = 0$ for every $w \in \mathfrak{g}$, that is $\text{ad}_{\text{curl } m}^* m = 0$; the conservation law then follows from (3) and $\langle a, \text{curl } b \rangle = \langle \text{curl } a, b \rangle$:

$$\frac{1}{2} \frac{d}{dt} H(m) = \langle \dot{m}, \text{curl } m \rangle = \langle \text{ad}_u^* m, \text{curl } m \rangle = \langle m, [u, \text{curl } m] \rangle = -\langle m, [\text{curl } m, u] \rangle = 0.$$

For the first claim write $\omega = \text{curl } m$ and $I = \int m_i (\omega_k \partial_k w_i - w_k \partial_k \omega_i)$. Integrating the first term by parts and using $\text{div } \omega = 0$ gives $\int m_i \omega_k \partial_k w_i = -\int (\omega_k \partial_k m_i) w_i$. For the second, $\int m_i w_k \partial_k \omega_i = \int w_k \partial_k (m_i \omega_i) - \int w_k \omega_i \partial_k m_i$, and the first piece is $-\int (\text{div } w)(m \cdot \omega) = 0$. Hence

$$I = \int w_k \omega_i (\partial_k m_i - \partial_i m_k) = \int w_k \omega_i \varepsilon_{kil} \omega_l = 0,$$

since $\partial_k m_i - \partial_i m_k = \varepsilon_{kil} \omega_l$ and ε_{kil} is antisymmetric in (i, l) . $\square$

The proof uses nothing about A ; it uses only that $\dot{m} = \text{ad}_v^* m$ for *some* v . That is exactly why the condition has content for a cascade, whose $\dot{m}$ need not be of that form.

Definition 3.2. For a bilinear operator C on $\mathfrak{g}$ and a positive definite A , the *helicity defect* is

$$D(X; A) := \langle A^{1/2} C(X, X), \text{curl}(A^{1/2} X) \rangle = \frac{1}{2} \frac{d}{dt} H(m) \Big|_{\dot{X}=C(X, X)}.$$

By Lemma 3.1, $D(X; A) \equiv 0$ whenever $C = B_A$.

4. THE REDUCTION TO DETERMINANTS

Lemma 4.1. For every symmetric 2×2 matrix S and $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ one has $SJS = \det(S) J$.

Proof. With $S = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, $SJ = \begin{pmatrix} b & -a \\ c & -b \end{pmatrix}$ and $SJS = \begin{pmatrix} 0 & b^2 - ac \\ ac - b^2 & 0 \end{pmatrix} = (ac - b^2)J$. $\square$

The identity is special to two dimensions; in higher dimensions SJS is not a multiple of J . That $k^\perp$ is two-dimensional is what makes the next proposition possible.

Proposition 4.2. Fix an orthonormal frame of $k^\perp$ for each pair, positively oriented with respect to k , and write $\hat{C}(k), \hat{X}(k) \in \mathbb{R}^2 \otimes \mathbb{C}$ in that frame. Then

$$D(X; A) = \sum_{k \in \mathcal{P}} \sqrt{\det A_k} c_k(X), \quad c_k(X) = -2|k| \text{Im} \langle \hat{C}(k), J \hat{X}(k) \rangle. \quad (4)$$

The inertia operator enters only through the numbers $\sqrt{\det A_k}$, one per leg pair, and enters linearly.

Proof. On $k^\perp$ the curl acts by $a \mapsto ik \times a$, and $k \times$ is $|k|$ times the quarter turn J in the chosen frame; equivalently $|k|Ja = k \times a$, so that $c_k(X) = -2 \text{Im} \langle \hat{C}(k), k \times \hat{X}(k) \rangle$ and neither a frame nor a normalisation is needed to evaluate it. Writing S_k for the matrix of $A_k^{1/2}$, the mode- k term of Definition 3.2 is $\langle S_k \hat{C}(k), i|k|JS_k \hat{X}(k) \rangle$, whose inner matrix is $S_k JS_k = \det(S_k)J$ by Lemma 4.1. Because X and $C(X, X)$ are real-valued vector fields, $\hat{C}(-k) = \hat{C}(k)$ and $\hat{X}(-k) = \hat{X}(k)$, so the terms at k and $-k$ are complex conjugates and their sum is twice a real part; the stated normalisation collects it. In particular $D(X; A)$ is real, as it must be. Finally $\det(S_k) = \det(A_k^{1/2}) = \sqrt{\det A_k}$. $\square$

Lemma 4.1 is available one step earlier than we have used it, before any time derivative is taken, and there it says something stronger.

Corollary 4.3. With $m = A^{1/2}X$ as in (2),

$$H(m) = \sum_{k \in \mathcal{P}} \sqrt{\det A_k} h_k(X), \quad h_k(X) = -2|k| \text{Im} \langle \hat{X}(k), J \hat{X}(k) \rangle, \quad (5)$$

which is (4) with $\hat{C}$ replaced by $\hat{X}$. The value of the momentum helicity, and not merely its conservation, is a linear functional of the numbers $\sqrt{\det A_k}$, with coefficients that do not see the inertia operator at all.

Proof. Verbatim the proof of Proposition 4.2 with $\hat{C}(k)$ replaced by $\hat{X}(k)$: the mode- k term of $\langle m, \text{curl} m \rangle$ is $\langle S_k \hat{X}(k), i|k|JS_k \hat{X}(k) \rangle$, whose inner matrix is $S_k JS_k = \det(S_k)J$, and the pairing of k with $-k$ is as before. The pairing is the Hermitian one; the bilinear form $v^\top J v$ vanishes identically for antisymmetric J , so no other reading of (5) is available. $\square$

Remark 4.4 (What the criterion sees, and what it cannot). Proposition 4.2 and Corollary 4.3 are the same statement from two sides: the entire Casimir content of the inertia operator is the vector

$$\delta(A) := (\sqrt{\det A_k})_{k \in \mathcal{P}} \in (0, \infty)^{\mathcal{P}}.$$

That is the precise sense in which the criterion is sharp, and it is equally a ceiling, because the fibres of δ are explicit. On each leg pair A_k is a symmetric positive definite form on the two-dimensional plane $k^\perp$ and so carries three real parameters, of which the determinant pins one; the ratio of its eigenvalues and the orientation of its principal axes inside $k^\perp$ survive. Everything in this paper — the defect D , the matrix $M(C)$, its corank, its null vector, and now the numerical value of the conserved helicity itself — is constant along those two directions per leg pair, while the operator they parametrise is not. Proposition 6.4 measures both halves of that sentence.

5. THE CRITERION

Fix a finite family $X_1, \dots, X_N$ of test fields and set $M(C) := [c_k(X_s)]_{s \leq N, k \in \mathcal{P}}$.

Theorem 5.1. *If $C = B_A$ for a symmetric positive definite A , then the vector $(\sqrt{\det A_k})_{k \in \mathcal{P}}$ is an entrywise positive null vector of $M(C)$. Consequently, if $M(C)$ has full column rank, or if every null vector of $M(C)$ has a non-positive entry, then C is not an Euler–Arnold operator, for any inertia operator whatsoever.*

Proof. By Lemma 3.1, $D(X; A) = 0$ for every X , so by Proposition 4.2 the vector $(\sqrt{\det A_k})_k$ annihilates every row of $M(C)$. It is entrywise positive because A_k is positive definite on a two-dimensional space, so $\det A_k > 0$. $\square$

The positivity hypothesis looks like a technical convenience. It is not: it is an exact transcription, and the criterion loses nothing by it.

Proposition 5.2 (the criterion is sharp for this Casimir). *Let C be any bilinear operator on $\mathfrak{g}$. The following are equivalent.*

- (1) *There is a symmetric positive definite A with $D(X; A) = 0$ for every X .*
- (2) *There is $d \in (0, \infty)^{\mathcal{P}}$ with $\sum_{k \in \mathcal{P}} d_k c_k(X) = 0$ for every X .*

Moreover, if (2) holds then (1) holds with $A_k = d_k I$ on each $k^\perp$; in particular an A that is a positive scalar on every leg plane is as good as any.

Proof. (1) $\Rightarrow$ (2) is Proposition 4.2 with $d_k = \sqrt{\det A_k} > 0$. For (2) $\Rightarrow$ (1), set $A_k := d_k I$ on $k^\perp$ and $A_{-k} := A_k$. This is symmetric and positive definite, and $\det A_k = d_k^2$, so $\sqrt{\det A_k} = d_k$ and $D(X; A) = \sum_k d_k c_k(X) = 0$ for every X by Proposition 4.2. $\square$

Remark 5.3. Two readings. First, the two clauses of Theorem 5.1 do not merely suffice to exclude C : on a test family rich enough that a vector annihilating every row annihilates $(c_k(X))_k$ for every X — call it *complete*, and note that the families used in §6 have more rows than columns — they fail precisely when some positive definite A has its momentum helicity conserved by C . So the criterion extracts everything this Casimir can give, and a cascade surviving it cannot be excluded by the momentum helicity at all.

Second, Proposition 4.2 says the inertia operator enters only through $\sqrt{\det A_k}$, so anisotropy of A is invisible here; Proposition 5.2 is the converse, that nothing is thereby lost, every admissible vector of determinants being realised by a leg-scalar A . The search over inertia operators is a search over one positive number per leg pair, and over nothing else.

The other structure the null space can have is a silent leg, and that too is exactly characterised.

Lemma 5.4 (silent legs). *Fix $k \in \mathcal{P}$. Then $c_k(X) = 0$ for every X if and only if $\hat{C}(k) = 0$ for every X with $\hat{X}(\pm k) = 0$. If moreover $2k \notin \mathcal{S}$, this holds if and only if $\hat{C}(k) = 0$ for every X — the cascade has no output at leg k .*

Proof. If $\hat{C}(k)$ vanishes on the stated fields, let Y be one of them, let Z be supported on $\{\pm k\}$, and put $X = Y + tZ$ with $t \in \mathbb{R}$. Since C is quadratic, $\hat{C}(k)[X] = \hat{C}(k)[Y] + 2t\hat{C}(k)(Y, Z) + t^2\hat{C}(k)[Z]$, while $\hat{X}(k) = t\hat{Z}(k)$, so $c_k(X) = -2|k|\operatorname{Im}\langle\hat{C}(k)[X], J\hat{X}(k)\rangle$ is a polynomial in t with no constant term. Conversely, suppose $c_k \equiv 0$ and take $X = Y + tZ$ as above. A real polynomial vanishing for all real t has vanishing coefficients, and the coefficient of t is $-2|k|\operatorname{Im}\langle\hat{C}(k)[Y], J\hat{Z}(k)\rangle$. As Z ranges over the real fields supported on $\{\pm k\}$, $\hat{Z}(k)$ ranges over all of $k^\perp \otimes \mathbb{C}$; taking $\hat{Z}(k) = J^{-1}e$ and $\hat{Z}(k) = iJ^{-1}e$ for e in a basis of $k^\perp$ returns the imaginary and the real part of the corresponding component of $\hat{C}(k)[Y]$, so $\hat{C}(k)[Y] = 0$.

For the last statement, let X be arbitrary and split $X = Y + Z$ with Z supported on $\{\pm k\}$ and $\hat{Y}(\pm k) = 0$. A contribution to $\hat{C}(k)$ comes from a pair $k' + l' = -k$ with $k', l' \in \mathcal{S}$. If both lie in $\{\pm k\}$ then $k' + l' \in \{2k, 0, -2k\}$, and $-k$ is none of these for $k \neq 0$; if exactly one does, the other is $-k \mp k \in \{-2k, 0\}$, and both are outside $\mathcal{S}$ by hypothesis and by mean-zerosness. Hence $\hat{C}(k)[X] = \hat{C}(k)[Y]$. $\square$

Remark 5.5. On the lattice $0 < |k|^2 \leq 4$ of Proposition 6.9 the hypothesis $2k \notin \mathcal{S}$ holds for every leg with $|k|^2 > 1$, that is for 13 of the 16 pairs; at the three with $|k|^2 = 1$ the lemma still gives the first equivalence, which is what the stratification uses.

Remark 5.6 (numerical check). Both statements are validated end to end on that lattice by `sharpcrit.py`. For a random positive d the leg-scalar $A_k = d_k I$ has $\sqrt{\det A_k} = d_k$ to 4.4×10^{-16} ; the helicity defect of Definition 3.2, evaluated directly and not through (4), is unchanged to 1.9×10^{-16} relative when that A is replaced by a genuinely anisotropic A' with the same leg determinants, and equals $\sum_k d_k c_k(X)$ with ratio 1.000000 and spread 3.0×10^{-14} across 22 test fields — which is Proposition 4.2 in the normalisation used here. For Lemma 5.4, muting the output at the leg $k = (-2, 0, 0)$, for which $2k \notin \mathcal{S}$, makes c_k vanish identically while the unmuted cascade has $|c_k| \sim 8 \times 10^2$; and fields supported on $\{\pm k\}$ alone contribute exactly zero to $\hat{C}(k)$, which is the splitting step of the proof.

Theorem 5.7 (locality). *Let C be a cascade operator on a mode set $\mathcal{S}$, finite or not, and let $\mathcal{S}_0 \subseteq \mathcal{S}$ be finite and closed under $k \mapsto -k$. If $M(C)$ computed on $\mathcal{S}_0$ has full rank, then C is not an Euler–Arnold operator on $\mathcal{S}$, for any positive definite A .*

Proof. Suppose $C = B_A$ on $\mathcal{S}$, and let X be supported on $\mathcal{S}_0$. By Proposition 4.2, $c_k(X)$ carries the factor $\hat{X}(k)$, so $c_k(X) = 0$ for every $k \notin \mathcal{S}_0$; and $\hat{C}(p)$ is determined by the $\mathcal{S}_0$ -modes alone, since the triad sum runs over $k + l = -p$ with $\hat{X}(k), \hat{X}(l)$ vanishing off $\mathcal{S}_0$. Hence $0 = D(X; A) = \sum_{k \in \mathcal{S}_0} \sqrt{\det A_k} c_k(X)$ for every such X , so $(\sqrt{\det A_k})_{k \in \mathcal{S}_0}$ is an entrywise positive null vector of $M(C)$ on $\mathcal{S}_0$, contradicting full rank. $\square$

So the finite computations of §6 are not a model of the question: they are certificates for it. Proposition 6.9 exhibits, in exact integer arithmetic, one window of 32 modes on which a cascade has full rank; by Theorem 5.7 that operator is excluded on the whole lattice. Combining with Theorem 8.1, and noting that restriction to a window is a surjective linear map, so that the preimage of a proper Zariski-closed set is again one:

Corollary 5.8. *Outside a proper Zariski-closed set of coefficients and profiles, no local cascade operator on the full lattice is an Euler–Arnold operator, for any positive definite inertia operator.*

Two features deserve emphasis. First, the criterion never refers to the species count, to the number of terms in a sum, or to invertibility of a frame matrix; it applies verbatim to sums of arbitrarily many cascade terms. Second, it fails in two distinct ways — full rank, and rank deficiency with no positive null vector — and both occur in practice, the second being the one that catches Tao’s own operator.

6. MEASUREMENTS

All computations below are rank and residual computations in fixed-size linear algebra; no optimiser or search is used at any point. Each is validated on data known to satisfy the hypothesis before being applied.

Proposition 6.1 (numerical). *The premise of Lemma 3.1 holds in the discretisation. Computing $\dot{m} = -P(u \cdot \nabla m + (\nabla u)^\top m)$ in real space on a 16^3 grid with spectral derivatives, $|\frac{d}{dt}H|/|H|$ is at most 1.5×10^{-14} for $A = I$, $|k|^{-2}I$, $|k|^{3/2}I$ and three genuinely anisotropic symbols of the form $a(|k|)P_k + b(|k|)P_k(n \otimes n)P_k$. Computing instead through (1) directly, the defect of Definition 3.2 is 2.5×10^{-16} relative.*

Proposition 6.2 (numerical). *Formula (4) is exact: over random cascades, random positive definite A and random test fields, the two sides of (4) agree to 8.4×10^{-16} relative.*

Proposition 6.3 (numerical). *On the mode set $|k| \leq 2$ with 16 leg pairs, and for four random anisotropic A_0 , the matrix $M(B_{A_0})$ has corank exactly one, its smallest singular value being 10^{-16} relative; the null vector agrees with $(\sqrt{\det A_{0,k}})_k$ to ten decimal places, and its smallest entry is between 0.04 and 0.19 of its largest. It agrees with $(\det A_{0,k})_k$ only to 0.972, so the computation distinguishes the two and selects the exponent predicted by Proposition 4.2.*

Proposition 6.4 (numerical). *Fix the base point A_0 of Proposition 6.3 and move it along the fibre of δ : at every leg pair, scale the eigenvalues of $A_{0,k}$ by (μ, μ^{-1}) and rotate its principal axes through θ inside $k^\perp$, which preserves positive definiteness and every determinant. Over the grid $\mu \in \{1, 2, 4\}$, $\theta \in \{0, \pi/4, \pi/2\}$ the determinants are held to 5.8×10^{-16} relative while $\|A - A_0\|/\|A_0\|$ reaches 1.03, the condition numbers of the A_k staying between 2.9 and 11.7, so that no near-singular inertia operator is doing the work. At every one of the nine points the corank of $M(B_A)$ is one, the residual of the true determinant vector is at most 3.5×10^{-16} , the null vector agrees with the one at A_0 to ten decimal places, and the value of $H(m)$ is reproduced to machine precision, as (5) requires. The analysis does not follow. The stretching production $P = \langle w, (w \cdot \nabla)u \rangle$ with $w = \text{curl } m$ — which by [4, Prop. 2.2] obeys the unmodified Euler stretching equation for every inertia operator, and whose time integrability is what the continuation criterion of that note requires — has root-mean-square over twenty test fields, normalised as $P/E^{3/2}$ with $E = \langle m, u \rangle$, ranging from 0.087 to 0.300 across the same grid, a factor of 3.46, against a sampling floor of 1.42 obtained by splitting the field set in half at fixed A_0 . The conserved energy itself moves by about 30%.*

Proposition 6.5 (numerical). *Formula (5) is exact. Against real-space evaluation of $\langle m, \text{curl } m \rangle$ on a 16^3 grid, over seven inertia operators spanning five distinct determinant vectors, with three test fields each, the two sides agree to 1.1×10^{-14} relative. In the velocity variable $m = Au$, where the same computation returns $\det A_k$ in place of $\sqrt{\det A_k}$, the agreement is 1.4×10^{-14} . The un-conjugated pairing returns zero to 3×10^{-15} at every point, confirming the reading fixed in the proof of Corollary 4.3.*

Proposition 6.6 (numerical). *Let C be a cascade operator with α admissible in the sense of Lemma 2.2 and random profiles. Then $\langle C(X, X), X \rangle$ vanishes to 10^{-17} , while $D(X; A)$ is of*

order 10^{-2} relative for every A tested, and $M(C)$ has full column rank. Over twelve samples at $m = 2, 3, 4$ the ratio $\sigma_{\min}/\sigma_{\max}$ of $M(C)$ lies between 0.047 and 0.22. Restricting the profiles to disjoint species supports, as in Tao's construction, the rank is still full over eight further samples at $m = 3, 4$, with $\sigma_{\min}/\sigma_{\max}$ between 4.0×10^{-3} and 9.2×10^{-2} .

Proposition 6.7 (numerical). *The conclusion is not an artefact of the mode set. On $|k| \leq 2$, $|k| \leq 2.5$ and $|k| \leq 3$ — 16, 40 and 61 leg pairs — and on the images of the first two under a random invertible linear map, which preserves every triad while changing every length, angle and perpendicular plane, the corank of $M(B_{A_0})$ is one in all five cases with null vector $(\sqrt{\det A_{0,k}})_k$ positive, and $M(C)$ has full rank in all five.*

Proposition 6.8 (numerical). *For Tao's own coefficients the criterion excludes by its second clause. On the frequency support only the four constants with $\mu = 0$ and species $\{1, 3, 4\}$ survive Paper I, Theorem 6.3, and among the orderings $\alpha_{1,4,3} = \alpha_{4,1,3} = 0$. His triads are disjoint, each mode lying in exactly one, so (4) decouples into three unknowns per triad. The cyclic symmetrisation of α vanishes exactly, as Lemma 2.2 requires; the rank of the resulting 3-column matrix is 2, and the null vector is $(0, 1, 0)$ — supported on the leg whose block is the vanishing one. Two of the three determinants are forced to zero, so no positive definite A exists.*

Proposition 6.9 (exact). *There is a cascade operator, on an explicit integer mode set, for which $M(C)$ has full rank over $\mathbb{Q}$, with no floating point involved. Take the integer lattice $0 < |k|^2 \leq 4$ (32 modes, 16 pairs, 426 ordered triads); on each pair an integer basis $b_1 = k \times e$, $b_2 = k \times b_1$ of $k^\perp$; integer profiles and Gaussian-integer test fields in that basis; and integer α with vanishing cyclic symmetrisation. By Proposition 4.2 in the form $c_k(X) = -2 \operatorname{Im} \langle \hat{C}(k), k \times \hat{X}(k) \rangle$, every entry of $M(C)$ is then an integer. For the data recorded in the accompanying code, Gaussian elimination over $\mathbb{Q}$ gives $\operatorname{rank} M(C) = 16$.*

Remark 6.10. Proposition 6.8 is a consistency check rather than news about Tao: his surviving coupling has a block that vanishes identically, and Paper II, Corollary 5.6 already excludes it on that ground. Its value is that his coefficients are exactly the kind of non-generic point at which the criterion might have gone blind, and it does not.

Proposition 6.11 (numerical). *No cascade in the exceptional locus of Theorem 8.1 escapes by the second clause of Theorem 5.1, at least generically. Fix profiles and a vector d of positive determinants. Since $\hat{C}$ is linear in α , the set $H(d) = \{\alpha : \sum_k d_k c_k(X) = 0 \text{ for all } X\}$ is a linear subspace of the admissible space, computable by rank. Let $N = \{\alpha : C \equiv 0\}$, the trivial directions. Over the tested profiles and several d , including $d = (1, \dots, 1)$, one finds $H(d) = N$ exactly: in the cyclic-free space both have dimension 20, and in the admissible space of Lemma 2.2, where (4.2) is imposed, both are zero. So every admissible α whose cascade conserves the momentum helicity for some positive definite A is the zero tensor.*

Proposition 6.12 (numerical). *The exceptional locus admits a complete description in the tested setting, and it does not shelter anything. Since c_k is linear in α , the matrix $M(\alpha)$ is a linear pencil and Z is its rank-deficiency locus, stratified by the support of the null vector. Write $N = \{\alpha : C \equiv 0\}$. Then:*

- (1) *A null vector supported on a single leg k means $c_k \equiv 0$, which by Lemma 5.4 happens exactly when the cascade has no output at leg k ; that part is proved and not measured. What is measured is the size: for generic profiles the locus has dimension 4 inside the 20-dimensional admissible space, and two legs can be silent at once only for $\alpha = 0$.*
- (2) *For every other support, $H(d) = N$. Exhaustively over all 16, 120 and 560 supports of sizes 1, 2, 3 with three coefficient choices each, and over 3000 dense random d : for*

- generic profiles $\dim H(d) = 0$ except at the single-leg supports; for disjoint species supports $\dim H(d)$ is larger, but every α it contains gives the zero operator, to 2×10^{-14} .
- (3) On a branch, $M(\alpha)$ has rank exactly 15 of 16, and its null space is one-dimensional and supported on that one leg; having a zero entry it is not entrywise positive, so the criterion fires.

Corollary 6.13. *In the tested setting no admissible cascade operator with $C \neq 0$ is an Euler–Arnold operator — not merely the generic ones. Off Z there is no null vector at all; on Z the null vector is a coordinate direction, which fails positivity; and the α admitting a positive null vector are exactly those with $C \equiv 0$.*

By Proposition 5.2 the last clause is not a statement about matrices at all, and it is worth isolating in the form it actually has, because that form carries no test fields, no window and no linear algebra.

Question 6.14. *Let C be a local cascade operator with admissible α and arbitrary profiles, on any mode set. If some symmetric positive definite A has its momentum helicity conserved by C , must $C \equiv 0$?*

An affirmative answer upgrades Corollary 6.13 to a theorem with no tested setting in it, and by Theorem 5.1 it would say that the momentum helicity separates the two classes outright rather than generically. The evidence is Propositions 6.11 and 6.12; what makes the question tractable-looking is that Proposition 5.2 has reduced the search over inertia operators to one positive number per leg pair, so the unknown is a positive vector and not a family of matrices, and Lemma 5.4 already settles every vector with a zero entry.

Remark 6.15. This is what happens at Tao’s own coefficients, Proposition 6.8: they lie on the branch belonging to the leg whose block vanishes, the null vector is that coordinate direction, and positivity fails. The mechanism there was not special to him.

Remark 6.16. Since $\dim H(d)$ is upper semicontinuous in (ψ, d) , the equality $H(d) = N$ at one point gives it on a Zariski-open dense set, by the argument of Theorem 8.1. What is not excluded is a non-generic pair (ψ, d) at which $H(d)$ jumps; the locus of such pairs is closed and we have not described it.

7. THE VISCOUS PROBLEM

Everything above concerns the ideal equation. The criterion in fact survives the addition of viscosity, unchanged and uniformly in the viscosity parameter, and we record that here.

Let the momentum evolve by

$$\dot{m} = \text{ad}_u^* m + \nu A \Delta u, \quad (6)$$

so that viscosity acts on the velocity as it does in Navier–Stokes; for $A = I$ the added term is $\nu \Delta u$. Since A is symmetric, $\langle A \Delta u, \text{curl } m \rangle = \langle \Delta u, A \text{curl } m \rangle$, and we write

$$V(X; A) := \langle A \Delta u, \text{curl } m \rangle = - \sum_k |k|^2 \langle \hat{m}(k), \widehat{\text{curl } m}(k) \rangle,$$

the second equality because $(A \Delta u)^\wedge(k) = -|k|^2 \hat{m}(k)$.

Proposition 7.1. *Let C be a bilinear operator and replace (6) by $\dot{m} = A^{1/2} C(X, X) + \nu A \Delta u$. Then*

$$\frac{1}{2} \frac{d}{dt} H(m) = D(X; A) + \nu V(X; A),$$

with D the helicity defect of Definition 3.2. The second term depends only on (X, A) and not on C , so it takes the same value for C and for B_A .

Proof. Immediate from $\langle \text{ad}_u^* m, \text{curl } m \rangle = 0$, which is the computation in Lemma 3.1, together with the symmetry of A . $\square$

Two consequences, of opposite character. The first is negative, and worth stating because the opposite is a natural guess: the viscous term is *common to both classes* and cancels from any comparison between them, so no separator can be built out of the dissipation. A candidate C carries the same $\nu A \Delta u$ that B_A does — this is how Tao’s construction is posed, as an averaged Navier–Stokes equation with the unmodified viscous term — and the difference of the two helicity balances is $D(X; A)$, exactly as in the inviscid case.

The second is the useful one.

Corollary 7.2. *The criterion of Theorem 5.1 is unchanged by the addition of viscosity. If $M(C)$ has full column rank, or if every null vector of $M(C)$ has a non-positive entry, then C is not the Euler–Arnold operator of any inertia operator, for every $\nu \geq 0$ simultaneously. In particular the exclusion does not degenerate as $\nu \rightarrow 0$.*

Proof. By Proposition 7.1 the quantity distinguishing C from B_A is $D(X; A)$, which is the same function as in the inviscid case and is what Proposition 4.2 reduces and $M(C)$ tabulates. The hypothesis of Theorem 5.1 is therefore the same hypothesis, and ν does not appear in it. $\square$

Proposition 7.3 (numerical). *On the mode set $|k| \leq 2$, for $A = I$ and one admissible four-species cascade, $\frac{d}{dt}H(m)$ takes the values*

ν	Euler–Arnold	cascade
1	-3.166347×10^1	-3.707165×10^3
10^{-2}	-3.166347×10^{-1}	-3.675818×10^3
10^{-4}	-3.166347×10^{-3}	-3.675504×10^3
10^{-6}	-3.166347×10^{-5}	-3.675501×10^3
0	0	-3.675501×10^3

The Euler–Arnold column is exactly proportional to ν ; the cascade column converges to $2D(X; A) \neq 0$. Three validations accompany the computation: the Euler–Arnold vertex gives nonlinear helicity production 10^{-17} , the cascade satisfies the energy identity to 2.5×10^{-17} , and for $A = I$ the quantity V agrees with $-\int \omega \cdot \text{curl } \omega$ to machine zero, which is the textbook helicity balance $\frac{d}{dt}H = -2\nu \int \omega \cdot \text{curl } \omega$.

Remark 7.4. In the vocabulary of turbulence, Proposition 7.3 says that a cascade operator exhibits *anomalous helicity production*: helicity is generated at a rate that survives the inviscid limit and has no viscous origin, whereas for an Euler–Arnold flow the rate is $O(\nu)$ and vanishes with it. This is a restatement of Definition 3.2 and not an independent invariant. We note also that Onsager-type rigidity for helicity constrains genuine weak solutions rather than operators, and so does not apply to a cascade operator directly; the statement here is about the operator class, not about a solution.

Remark 7.5. The placement of the viscosity in (6) is a modelling choice. Taking $\nu \Delta m$ in place of $\nu A \Delta u$ changes V , but not the fact that the added term is independent of C , so Proposition 7.1 and Corollary 7.2 are unaffected. The numerical values in Proposition 7.3 are for one field and one coefficient tensor on a small mode set and are illustrative; what is not normalisation-dependent is the vanishing of the Euler–Arnold column with ν and the non-vanishing of the cascade limit.

8. WHAT THIS ESTABLISHES

Theorem 8.1. *Fix the integer mode set and the test fields of Proposition 6.9, and let the species count be four. Then there is a proper Zariski-closed subset Z of the admissible pairs (α, ψ) such that for every $(\alpha, \psi) \notin Z$ the associated cascade operator is not an Euler–Arnold operator, for any positive definite inertia operator.*

Proof. For fixed test fields, $\hat{C}(k)$ is bilinear in the profile pairings and linear in α , so each $c_k(X_s)$ is a polynomial in (α, ψ) . Full rank is the non-vanishing of some maximal minor, a polynomial condition, so $Z := \{\text{rank } M(C) < 16\}$ is Zariski-closed. It is proper because Proposition 6.9 exhibits a point outside it, by an exact computation over $\mathbb{Q}$. Off Z the matrix $M(C)$ has full rank, so it has no null vector at all, and Theorem 5.1 applies. $\square$

The witness is exact, so Theorem 8.1 carries no numerical input at all: the negative answer for generic cascade operators is a theorem, not a statement about a sample, and unlike the dimension counts of Paper III it does not degrade with the species count.

What is *not* established is threefold, and we state it plainly.

First, the exceptional locus of Theorem 8.1 is inhabited: Proposition 6.8 exhibits a point of it, where the criterion still excluded, by positivity rather than by rank. A cascade with both a rank deficiency and an entrywise positive null vector would escape the criterion entirely, and Propositions 6.11 and 6.12 say there is none: the α admitting a positive null vector are exactly those giving the zero operator. Part of that is now proved rather than measured. Every null vector with a zero entry is settled by Lemma 5.4, and by Proposition 5.2 what remains — a strictly positive null vector — is exactly the existence of a positive definite inertia operator whose momentum helicity the cascade conserves, which is Question 6.14. That residue is what was checked on two profile families and one window, and it is the only thing in the stratification that was.

Second, every computation reported here abstracts Tao’s species balls B_i to their centres, so that each species contributes one wavevector and one profile direction. This is the same abstraction under which Paper I–Paper III operate, but it is an abstraction: his ψ_i genuinely span balls of positive radius.

Third, the mode sets are finite, but by Theorem 5.7 this is no longer a restriction on the conclusion: a full-rank window certifies the operator on any ambient set containing it, the full lattice included. What is not addressed is a cascade whose profiles are not of the Fourier form of Definition 2.1, and the continuum limit in which the number of species or the annulus geometry is allowed to vary with the scale.

Remark 8.2 (The class the criterion lives in). Theorem 5.1 is a statement about Fourier symbols: $M(C)$ is assembled from $\hat{C}(k)$, and Proposition 4.2 reduces the defect $D(X; A)$ to determinants in the symbol of A . The criterion therefore speaks about translation-covariant inertia operators, and that is exactly the class in which the question behind Paper I–Paper III is still open. Tao [11] settles the unrestricted version, and settles it the other way: replacing the vector potential operator of the Euler equation by an arbitrary positive definite operator of order -2 leaves the bracket, the energy, the Kelvin circulation theorem, helicity and vorticity transport intact — the equation remains the geodesic flow on $\text{SDiff}(M)$ for a right-invariant metric — and yet admits smooth compactly supported data blowing up in finite time, with the metric within ε of the Euler metric. The operators he constructs are inhomogeneous and non-isotropic, and his construction fixes the order of A at that of Euler rather than varying it. So the unrestricted Euler–Arnold class is compatible with blowup, while the translation-covariant subclass is not

known to be, and it is that subclass which the separator proved here constrains. See Paper I, Section 7.

Remark 8.3 (The ceiling). Remark 4.4 and Proposition 6.4 bound what an argument of this kind can deliver, and the bound belongs next to the result rather than after it. Two Euler–Arnold operators lying in one fibre of δ agree on the criterion exactly — same corank, same null vector to ten decimal places — and agree on the numerical value of the conserved momentum helicity as well, yet differ by a factor of several in the stretching production whose integrability the continuation criterion of [4] asks for. Casimir data therefore does not determine the analysis, and no refinement of the criterion reaches inside the fibre: what separates cascade operators from Euler–Arnold operators is exactly the information δ retains, and the two remaining parameters per leg pair lie outside the reach of the method. Read together with Remark 8.2 this says where the separation does and does not point. It settles the classification question Paper III handed on, and it is not a step towards regularity, for the same reason that [11] is not obstructed by any of the invariants preserved here.

Remark 8.4. The contrast with Paper III is worth recording. There the two classes were separated by a codimension that grew quadratically in the number of legs and still proved nothing, because the images might meet in special position. Here the separation is a rank, and a rank is not a dimension count: full rank is an open condition, it is witnessed by one evaluation, and Proposition 8.1 converts the witness into a statement about almost every cascade. The algebraic geometry that Paper III asked for turns out to be one line of it.

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The code underlying the computational statements is available from the author.

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